Magnetosphere and changing fields

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use evidence to distinguish a magnetic-field model from an unsupported space-weather claim.

SUCCESS CRITERIA - I can...

1. Connect changing magnetic fields with induced electric effects.
2. Distinguish an assumption from an observation.
3. Name evidence required for a space-weather prediction.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

At fixed current, move from r to 3 r from a long straight wire. Using B proportional to 1/r, compare field strength.

Marzano check - circle: ① ② ③

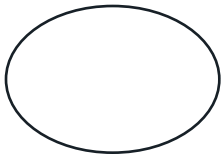
02 Worked example: explain the model

Reference: $|\text{emf}_{avg}| = N|\Delta\Phi|/\Delta t$; $\Phi = BA$.
Uniform perpendicular field for $\Phi = BA$. Use supplied/measured field changes; an asteroid alone is not evidence of an electromagnetic pulse.

Worked reasoning: A changing magnetic environment can induce voltages in conducting loops. A static magnetic field alone is not the same as a changing flux through a particular circuit.

Your turn: Sketch a conducting loop and field arrows; show one way flux can change.

Label field direction, loop area and what changes.



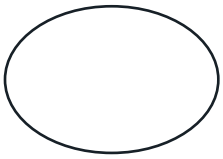
Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: ① ② ③

03 Read and interpret the model

Draw a coil in a changing magnetic field. Label what changes and where an induced emf could appear.

Label field direction, loop area and what changes.



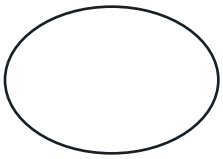
Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: ① ② ③

04 Apply the physics

A coil simulation produces 2 V for one magnetic-field change rate. Double that rate while keeping coil area, orientation and turns fixed. Predict the new emf and label what changes in the flux model.

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: ① ② ③

05 Explain your reasoning

A headline claims an approaching rock will disable power grids through an EMP, but gives no field measurements or mechanism. Name two pieces of evidence needed before accepting it.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Explain what a potential difference of 6 V means in joules per coulomb.
Label voltage, charge and energy with their distinct units.

Initial / supply _____

V: ____ J/C

Final / component _____

V: ____ J/C

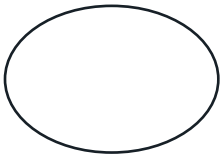
Charge: ____ C Energy transfer: ____ J

Marzano check - circle: 1 2 3

07 Exit check

Does a constant flux through a stationary coil create sustained induced emf by Faraday's law?

Label field direction, loop area and what changes.



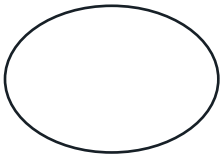
Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Compare a small loop and a large loop with the same orientation in the same uniformly changing field. Predict which may have greater induced emf and state the assumptions.

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: 1 2 3